

Post-Training SVD Compression for DoRA

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DoRA-SVD: Motivation

Goal

Compress a trained DoRA adapter by pruning redundant singular components from the direction update, while keeping validation performance close to rank-16 DoRA.

- ▶ DoRA separates the adapter update into magnitude and direction.
- ▶ The proposed compression only modifies the LoRA-style direction branch.
- ▶ The DoRA magnitude parameter is not pruned.

$$W'_\ell = m_\ell \cdot \frac{W_\ell + \Delta W_\ell}{\|W_\ell + \Delta W_\ell\|}, \quad \Delta W_\ell = B_\ell A_\ell$$

Post-Training Direction SVD

- ▶ First train a standard rank-16 DoRA adapter.
- ▶ After training, compute the effective direction update for each adapted layer.
- ▶ Apply SVD to the direction update, not to the full DoRA-scaled weight.

$$\Delta W_\ell = B_\ell A_\ell = U_\ell \text{diag}(\sigma_\ell) V_\ell^\top$$

Energy-based rank selection

Choose the smallest k_ℓ such that the first k_ℓ singular components preserve the target energy.

$$\frac{\sigma_{\ell,1}^2 + \cdots + \sigma_{\ell,k_\ell}^2}{\sigma_{\ell,1}^2 + \cdots + \sigma_{\ell,16}^2} \geq \tau, \quad \tau \in \{0.90, 0.95\}$$

Compact Adapter Export

- ▶ The pruned update is reconstructed from the retained singular components.
- ▶ The reconstructed update is factorized back into compact LoRA matrices.
- ▶ Each layer can have a different effective rank k_ℓ .

$$\widehat{\Delta W}_\ell = U_{\ell,k} \text{diag}(\sigma_{\ell,k}) V_{\ell,k}^\top = \widehat{B}_\ell \widehat{A}_\ell$$

$$\widehat{A}_\ell \in \mathbb{R}^{k_\ell \times d_{\text{in}}}, \quad \widehat{B}_\ell \in \mathbb{R}^{d_{\text{out}} \times k_\ell}$$

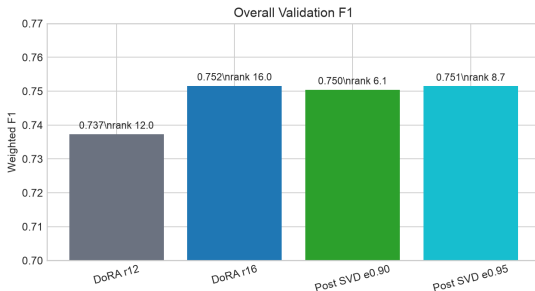
$$\#\theta_\ell^{\text{LoRA}} = k_\ell(d_{\text{in}} + d_{\text{out}}) < 16(d_{\text{in}} + d_{\text{out}})$$

DoRA-SVD Experimental Setup

Item	Setting
Datasets	SST-2, IMDb
Label noise	Clean, 20%, 40%
Data sizes	1,000 and 5,000 training examples
Seeds	13, 21, 42
Model	ModernBERT-based sequence classifier
Baselines	DoRA rank 12, DoRA rank 16
Compression	DoRA rank 16 + post-training SVD energy 0.90 / 0.95
Total runs	$2 \times 3 \times 2 \times 4 \times 3 = 144$

Main Result: Similar F1 with Lower Rank

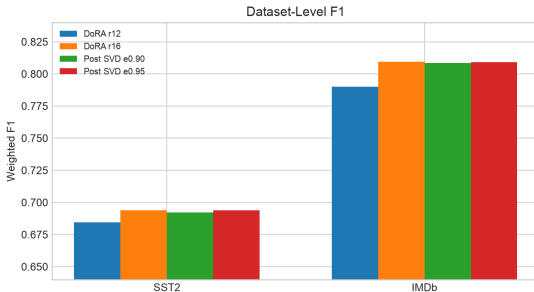
- ▶ Energy 0.95 preserves almost the same validation F1 as rank-16 DoRA.
- ▶ Average effective rank decreases from 16 to 8.68.



$$\text{DoRA}_{16} : F1 = 0.7516, r = 16 \quad \Rightarrow \quad \text{SVD}_{0.95} : F1 = 0.7515, \bar{r} = 8.68$$

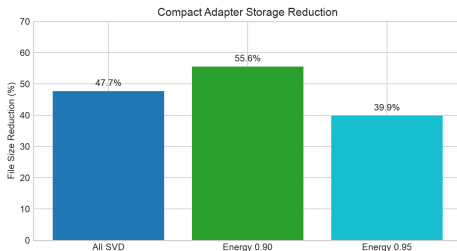
Dataset-Level Result

- ▶ The result is consistent across SST-2 and IMDB.
- ▶ This suggests the compression is not only fitting one dataset.



Compact Adapter Size Reduction

- ▶ Compact export stores only retained direction components.
- ▶ Energy 0.90 gives stronger compression; energy 0.95 is more conservative.



$\tau = 0.90$: file ↓ 55.6%, $\tau = 0.95$: file ↓ 39.9%

Compact Validation

Validation check

The compact adapter is loaded and compared against the full-shape pruned adapter. A zero difference means the compact file represents the same adapter function.

$$\hat{B}_\ell \hat{A}_\ell \equiv \text{zeroPad}(\hat{B}_\ell) \text{zeroPad}(\hat{A}_\ell)$$

Metric	Value
Validated compact adapters	72 / 72
Max compact-vs-full F1 difference	0.0
Max tensor reconstruction difference	0.0

DoRA-SVD Takeaway

Main finding

Post-training SVD reveals that trained DoRA direction updates contain redundant rank components.

- ▶ Rank-16 DoRA can be compressed to average rank 8.68 with energy 0.95.
- ▶ Validation F1 remains nearly unchanged: $0.7516 \rightarrow 0.7515$.
- ▶ Compact export reduces actual adapter file size by about 39.9%.
- ▶ The magnitude branch remains untouched, so the method specifically targets the direction update.